B.Comp. Dissertation

Advanced Tokenization Methods for Molecular Foundation Models

By

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Department of Computer Science

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Abstract

A chemical language model reads a molecule as a sequence of tokens. Its tokenizer determines whether a token represents a character, an atom, or a larger molecular fragment. This project compares tokenization methods at these different scales through language model pretraining and molecular property prediction. In the initial benchmark, a glyph-based tokenizer achieved the highest mean test score on both the HIV activity and blood-brain barrier penetration tasks.

The fragment experiments examine two choices: where to cut a molecule and how to record the connections between its fragments. Merging frequently adjacent fragments shortened tokenized sequences, but increased the proportion of unknown tokens and produced mixed prediction results. With the fragment boundaries held fixed, storing attachment sites separately allowed different attachment patterns of the same core to share one vocabulary entry. On both tasks, the mean test score of this enhanced TSDY representation was higher than that of the original representation with attachment sites embedded in the fragment words. These findings show why sequence length and vocabulary coverage should be evaluated alongside predictive performance.

Subject Descriptors:

10010147.10010257 Computing methodologies → Machine learning
10010405.10010444.10010450 Applied computing → Life and medical sciences → Bioinformatics

Keywords:

Molecular tokenization, molecular representations, chemical language models, molecular property prediction.

Implementation Software and Hardware:

Python, RDKit, PyTorch, Hugging Face Transformers, NUS School of Computing cluster (AMD EPYC 7763 CPU and NVIDIA H100 GPU).

Contents

Abstract i
1 Introduction 1
1.1 Background and motivation 1
1.2 Objectives and research questions 2
1.3 Contributions and scope 2
2 Background and related work 3
2.1 Molecular representations and language models 3
2.2 Character, glyph, and atom-based tokenization 3
2.3 Fragment-based molecular representations 5
2.4 Data-driven fragment extraction 7
3 System design and implementation 8
3.1 Benchmark framework and design objectives 8
3.2 Shared fragmentation and principal-subgraph merging 9
3.3 Enhanced TSDY 10
3.4 Reproducibility and implementation checks 13
4 Experimental methodology 14
4.1 Benchmark comparisons 14
4.2 Vocabulary construction and pretraining 14
4.3 Downstream evaluation 15
4.4 Intrinsic evaluation 15
5 Results and discussion 17
5.1 Initial benchmark of existing tokenizers 17
5.2 Shared fragmentation and PS merging 18
5.3 Enhanced TSDY on HIV and BBBP 18
5.4 Sequence length and vocabulary coverage 19
5.5 Discussion 20
6 Conclusions 21
6.1 Summary of findings 21
6.2 Limitations 21
6.3 Recommendations for future work 21
References 22
Appendix A: Declaration of AI assistance 24
Appendix B: Reproducibility details 25
Appendix C: Downstream results by configuration 26
Appendix D: Intrinsic results by configuration 27

1 Introduction

1.1 Background and motivation

Drawing inspiration from advances in natural language processing, researchers have developed chemical language models that learn from molecular structures represented as sequences (Sultan et al., 2024). A widely used representation is the Simplified Molecular Input Line Entry System (SMILES), which expresses a molecular graph as a string containing atom symbols and information about bonds, branches, rings, and stereochemistry (Weininger, 1988). Models such as ChemBERTa use self-supervised pretraining on molecular strings before being adapted to property prediction tasks (Chithrananda et al., 2020).

Before a model can process a molecular string, a tokenizer divides it into units and maps those units to entries in a vocabulary. Different tokenization schemes can divide the same notation differently, as illustrated in Figure 1. These choices determine which structural and syntactic information is grouped into a single token.

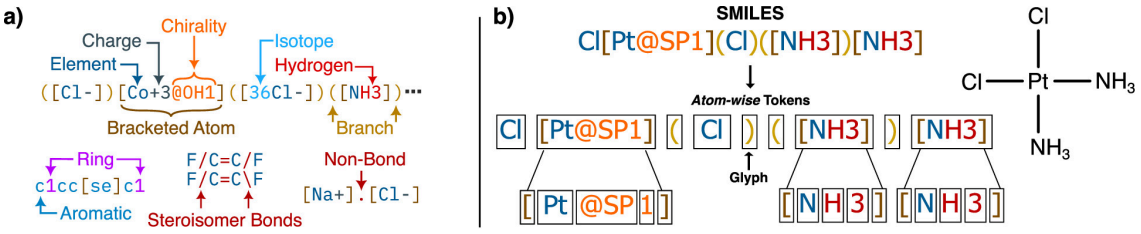


Figure 1: SMILES notation and tokenization granularity. Panel (a) shows structural features and panel (b) compares atom-wise and glyph-based tokenization. Source: (Wadell et al., 2026).

This project focuses particularly on fragment-based tokenization, in which a molecule is divided into connected groups of atoms before being represented as a sequence. Two design choices are involved: which bonds are cut to form fragments, and how the fragments and their connections are written in the sequence. Existing approaches include t-SMILES, a family of tree-based molecular representations (Wu et al., 2024a; Wu et al., 2024b), and fragSMILES, a fragment-based molecular representation (Mastrolorito et al., 2025).

A central question is how to represent a fragment independently of its connections to neighbouring fragments. If attachment annotations are included inside a fragment's vocabulary entry, the same underlying core can require different entries depending on its attachment positions or number of attachments. Keeping connection information separate may allow those forms to share a vocabulary entry. However, the connection information must still be represented if the original molecule is to be reconstructed, and this introduces additional syntax.

There is also a trade-off between grouping atoms into larger fragments and covering unfamiliar molecules with a finite vocabulary. In a tokenizer that assigns one token to each fragment, an unseen fragment is mapped to an unknown token unless another encoding mechanism is provided. Larger fragments may shorten sequences without improving coverage or predictive performance. These considerations motivate evaluating sequence length, unknown-token rates, and downstream prediction together.

1.2 Objectives and research questions

The objective is to compare molecular tokenization methods and investigate whether fragment-based representations can share vocabulary entries more effectively while retaining the information needed for reconstruction. The evaluation follows the pipeline in Figure 2: construct a tokenizer, pretrain a chemical language model, and finetune the model for molecular property prediction.

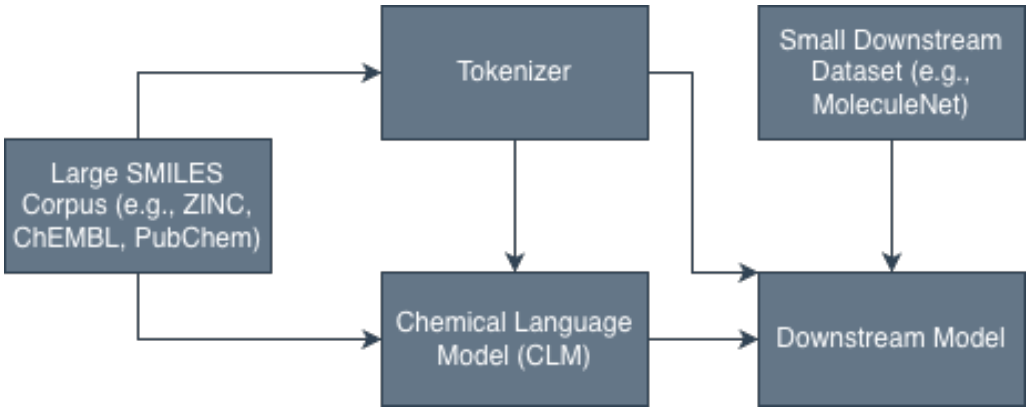


Figure 2: Conceptual evaluation pipeline. Corpus names are illustrative; the methodology chapter specifies the datasets and splits used.

The project addresses three research questions:

1. How do the choice of fragmentation rules and the merging of adjacent fragments affect sequence length, vocabulary coverage, and molecular property prediction?
2. Can separating fragment cores from attachment information improve vocabulary sharing when the fragment partition is held fixed?
3. To what extent do improvements in coverage or compression correspond to improvements in downstream predictive performance?

1.3 Contributions and scope

The project makes three practical contributions:

1. **A controlled fragmentation framework.** Different sequence representations can use the same molecular partition, making their assembly schemes easier to compare. The framework also supports learned merging of adjacent fragments, building on principal-subgraph mining (Kong et al., 2022).
2. **A representation with separate fragment and attachment information.** Building on t-SMILES (Wu et al., 2024a), the implementation represents clean fragment cores separately from the information needed to reconnect them and preserve supported stereochemistry. The design chapter introduces the particular t-SMILES variant and the enhanced encoding syntax.
3. **An evaluation of representation trade-offs.** The experiments combine property prediction with measurements of sequence length, unknown-token rates, and fragment vocabulary sharing. This allows improvements in one measure to be considered alongside their costs and limitations in others.

2 Background and related work

2.1 Molecular representations and language models

2.1.1 From molecular graphs to sequences

A molecule can be described as a graph whose vertices represent atoms and whose edges represent bonds. Sequence-based approaches express this structure as a string so that it can be processed by a language model. SMILES uses atom symbols together with branch, bond, ring-closure, and stereochemical notation (Weininger, 1988). Consequently, adjacent characters are not necessarily separate atoms, and graph neighbours need not appear next to each other in the string.

SELFIES is an alternative representation designed to produce valid molecular structures from sequences under its semantic rules (Krenn et al., 2020). This addresses a problem that is particularly relevant to molecular generation: whether a generated string can be interpreted as a valid molecule. SMILES and SELFIES thus differ in how their syntax constrains the mapping from strings to molecular structures.

2.1.2 Pretraining and property prediction

ChemBERTa applies the RoBERTa architecture to molecular strings (Chithrananda et al., 2020; Liu et al., 2019). In masked language modeling (MLM), selected input tokens are masked, and the model predicts their original identities using the surrounding sequence. This provides a training objective that does not require experimentally measured property labels. Through this objective, the model learns contextual relationships between tokens in molecular strings.

A pretrained model can subsequently be finetuned on labelled molecules for property prediction. ChemBERTa-2 also investigates multi-task regression, in which pretraining predicts computed molecular properties rather than masked tokens (Ahmad et al., 2022). MLM learns from token context, whereas multi-task regression provides supervision through computed properties.

2.2 Character, glyph, and atom-based tokenization

2.2.1 Statistical subword methods

Character-level tokenization uses small units, while subword methods combine units into longer recurring sequences. Byte-pair encoding (BPE) repeatedly merges frequent adjacent pairs (Gage, 1994). The initial units matter: starting from individual characters produces a different search space from starting with atom-level units. A learned sequence token is not necessarily a connected molecular fragment, since its boundaries are determined in the string representation.

The Unigram language-model approach instead learns a probabilistic vocabulary of candidate segments and prunes entries during training (Kudo, 2018). A further alternative, the Partition Cover Approach to Tokenization (PCATT), formulates vocabulary selection as a coverage problem and uses a greedy selection procedure (Lim et al., 2025).

2.2.2 Glyph-based tokenization

Character-level tokenization separates a string into individual characters without recognizing multi-character element symbols. For example, the characters `Sc` occur within `[Sc]`, denoting scandium, but also at the start of `Sc1ccccc1`, where `S` denotes sulfur and `c` an aromatic carbon. The surrounding syntax distinguishes these cases; the issue is not ambiguity in a complete valid SMILES string, but the loss of element boundaries when the notation is divided into characters.

SMIRK uses chemically meaningful glyphs to preserve such boundaries while decomposing bracketed atom specifications into reusable components (Wadell et al., 2026). Thus, `Sc` inside `[Sc]` is an element glyph, whereas the sulfur and aromatic carbon in `Sc1ccccc1` remain separate units. Compared with atom-wise tokenization, glyph-based tokenization also separates an atom's element from annotations such as charge or hydrogen count. Figure 1 illustrates this difference between atom-wise and glyph-based segmentation.

BPE can then be applied to the glyph sequence to learn frequently occurring combinations. Its pair-merging procedure remains the same, but it starts from glyphs rather than individual characters, preserving the initial glyph boundaries within the merged tokens.

2.2.3 Atom-based pair encoding

SMILES Pair Encoding (SPE) applies a BPE-style procedure to an atom-tokenized SMILES sequence: frequent adjacent token pairs are merged into progressively larger units (Li & Fourches, 2021). Atom Pair Encoding (APE) similarly learns pair merges from atom-based units (Leon et al., 2024). Unlike character- or glyph-based segmentation, this starting representation keeps bracketed atom expressions together rather than splitting their internal annotations. In all three settings, pair merging operates on a sequence; the difference lies in the units supplied to the algorithm. Figure 3 illustrates how learned sequence units differ from the initial atom-wise segmentation.

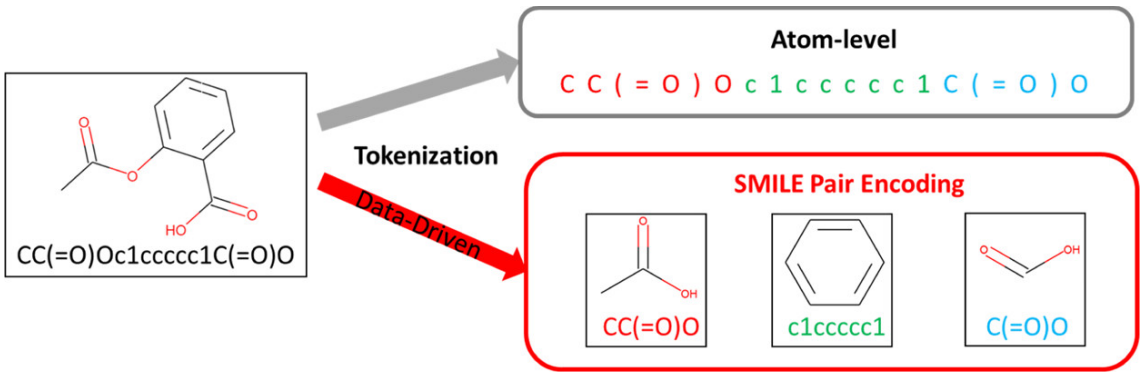


Figure 3: Atom-level segmentation and SMILES Pair Encoding. Source: (Li & Fourches, 2021).

Preserving atom boundaries does not make string-based merging equivalent to graph-based merging. Branches and ring closures can separate connected atoms in the serialized string. This motivates methods that partition the molecular graph before constructing the sequence, rather than relying only on adjacent sequence units.

2.3 Fragment-based molecular representations

2.3.1 Partitioning and serialization

Fragment-based representations introduce two distinct operations. First, a fragmentation rule divides the graph into substructures. Second, a serialization scheme records those substructures and their relationships as a sequence. Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) uses chemically defined bond-cleavage rules to construct fragment spaces (Degen et al., 2008). The t-SMILES framework supports BRICS and other fragmentation methods, including matched molecular pair analysis (MMPA) (Wu et al., 2024a). The resulting fragment distribution can therefore change even when the serialization scheme is held constant.

After serialization, the molecular string must be divided into tokens for the language model. One approach assigns a single token to each complete fragment string. In this case, different attachment annotations produce distinct vocabulary entries, even when the underlying fragment core is the same. Alternatively, subword tokenization divides fragment strings into smaller units that can be shared across different fragments.

2.3.2 SAFE

Sequential Attachment-based Fragment Embedding (SAFE) groups molecular notation into fragment blocks separated by dots and uses ring-closure notation to preserve connections between blocks (Noutahi et al., 2024). The fragment arrangement makes the substructures more explicit than in a conventional SMILES traversal. Figure 4 illustrates the representation.

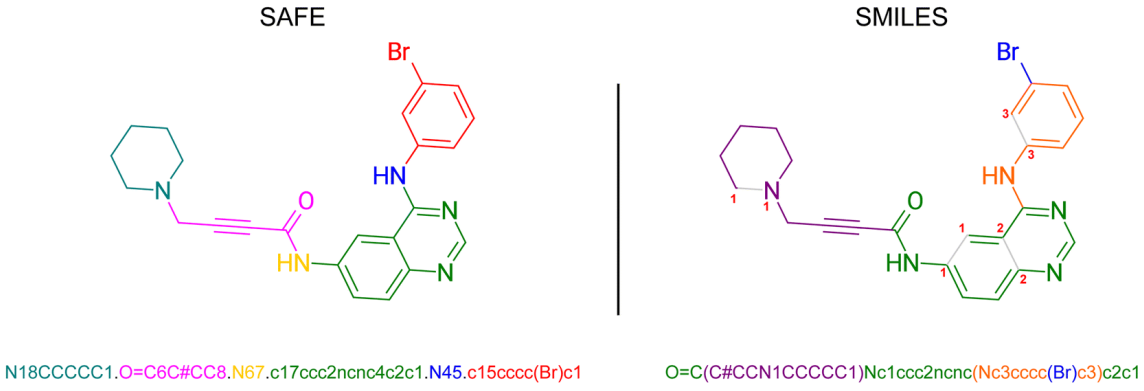


Figure 4: Fragment blocks in SAFE. Source: (Noutahi et al., 2024).

In SAFE, the connection symbols occur within the fragment blocks. Consequently, a fragment core can appear in different annotated strings depending on its connections. Subword tokenization can share portions of those strings, while whole-block tokenization treats each distinct block as a separate vocabulary item.

2.3.3 t-SMILES and its variants

The t-SMILES framework represents fragment relationships using a tree structure. Its original variants include TSSA, which uses shared atoms; TSDY, which uses dummy atoms without connection identifiers; and TSID, which uses dummy atoms with identifiers (Wu et al., 2024a). In the full-binary-tree representation, & denotes an empty tree node and ^ separates fragment strings.

The later TSIS representation simplifies TSID by reading from the acyclic molecular tree rather than the full binary tree. Related variants investigate alternative traversal orders (Wu et al., 2024b). Figure 5 shows the relationship between the fragment graph and these tree representations.

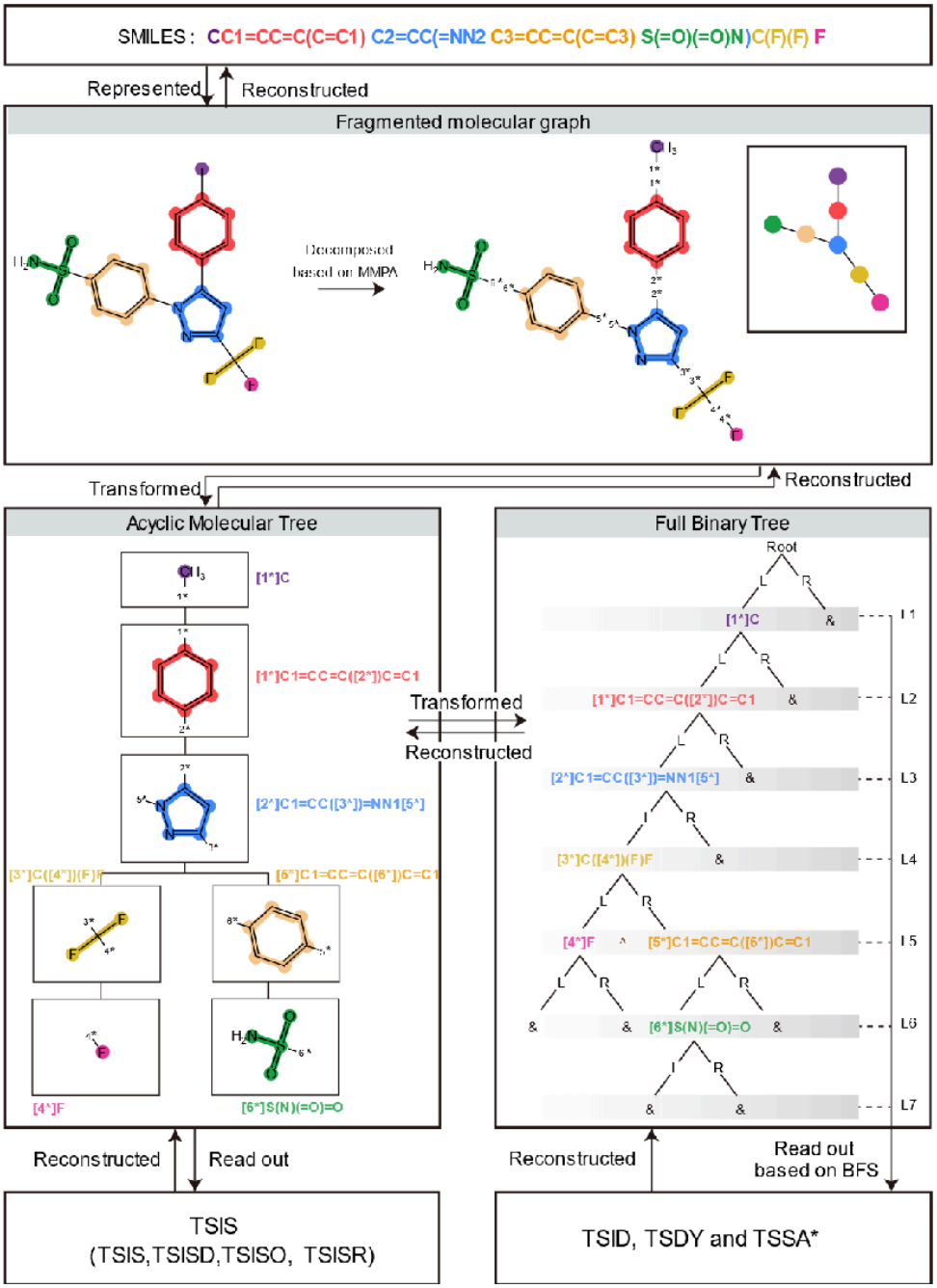


Figure 5: Fragmentation and tree representations used by t-SMILES. Source: (Wu et al., 2024b).

Dummy atoms, written as *, mark attachment sites within a fragment. TSID and TSIS additionally label these sites with connection identifiers (e.g., `[1*]`, `[2*]`), allowing corresponding sites in different fragments to be paired during reconstruction. TSDY retains the dummy atoms but omits these identifiers, so the explicit pairing information is lost.

Even without identifiers, fragments containing the same core can differ in the positions or number of dummy atoms. Under whole-fragment tokenization, these different strings require separate vocabulary entries. This motivates separating attachment information from the fragment core.

2.3.4 fragSMILES

fragSMILES represents fragments together with external connection annotations, providing a reference for separating fragment content from assembly information (Mastrolorito et al., 2025). Its default fragmentation strategy cuts single bonds that are not part of any ring. The sequence is organized around a main path in the fragment graph, with branches recorded recursively. Figure 6 illustrates the progression from an atom graph to fragments and a textual representation.

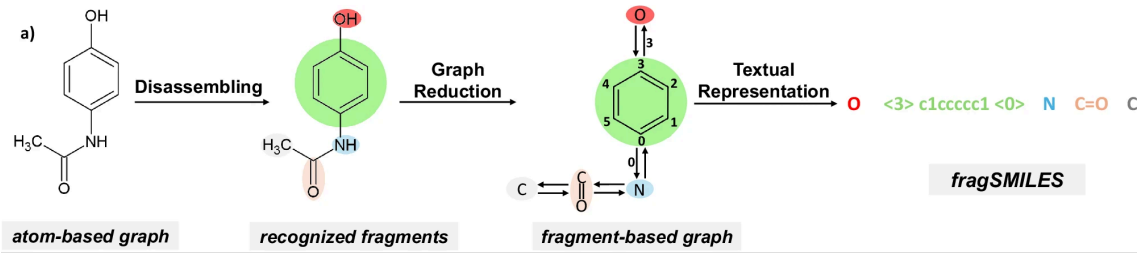


Figure 6: Fragment-based graph construction and connection annotations in fragSMILES. Source: (Mastrolorito et al., 2025).

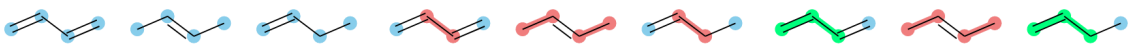
External connection annotations allow fragment content to be represented separately from its position in the assembled molecule. This offers a different balance from embedding dummy atoms and their identifiers inside fragment strings: the fragment core can be reused across attachment patterns, while separate syntax carries the assembly information.

2.4 Data-driven fragment extraction

2.4.1 Principal-subgraph mining

Principal-subgraph mining learns recurring molecular substructures from a corpus. This mining procedure was combined with a variational autoencoder and an assembly mechanism in PS-VAE (Kong et al., 2022). The mining procedure repeatedly merges frequently occurring adjacent subgraphs. Conceptually, it resembles pair merging in sequence tokenization, but adjacency is defined on the molecular graph.

As illustrated in Figure 7, the original mining procedure starts with individual atoms and repeatedly considers unions of neighbouring fragments. Candidate unions are counted across the corpus, and the most frequent pattern is added to the vocabulary. Its occurrences are merged before the next iteration (Kong et al., 2022).



(a) Initialization (b) Iteration 1 (c) Iteration 2

Figure 7: Principal-subgraph mining on a toy molecular dataset. Panels (a) to (c) show atom initialization and two successive merges. Colours indicate the fragment partition. Reproduced from Kong's explanation of PS-VAE (Kong, 2022).

3 System design and implementation

3.1 Benchmark framework and design objectives

3.1.1 Initial benchmark pipeline

The benchmark compares molecular tokenizers using a common RoBERTa architecture. It includes character- and glyph-based methods (BPE, PCATT, and SMIRK configurations), atom-based methods (SPE and APE), and fragment-word tokenizers for t-SMILES and fragSMILES. Selected configurations also use SAFE as the input representation. The t-SMILES configurations use either BRICS or MMPA fragmentation.

Each run has three stages:

1. **Vocabulary construction:** fit the tokenizer to the selected molecular corpus, or load an existing vocabulary. The APE baseline uses the authors' pretrained vocabulary.
2. **Masked language model (MLM) pretraining:** convert molecules to token IDs and train a RoBERTa encoder to predict masked tokens.
3. **Property prediction:** finetune the encoder for binary classification on two MoleculeNet datasets: HIV, which measures activity against the human immunodeficiency virus, and BBBP, which labels compounds by their ability to cross the blood-brain barrier (Wu et al., 2018).

3.1.2 Controlled fragment comparisons

Comparing BRICS-based t-SMILES with default fragSMILES changes both the bonds cut and the notation used to reconnect the fragments. The controlled benchmark separates these choices. As shown in Figure 8, each serializer receives either the same exocyclic partition or that partition after applying a fixed set of principal-subgraph (PS) merge rules.

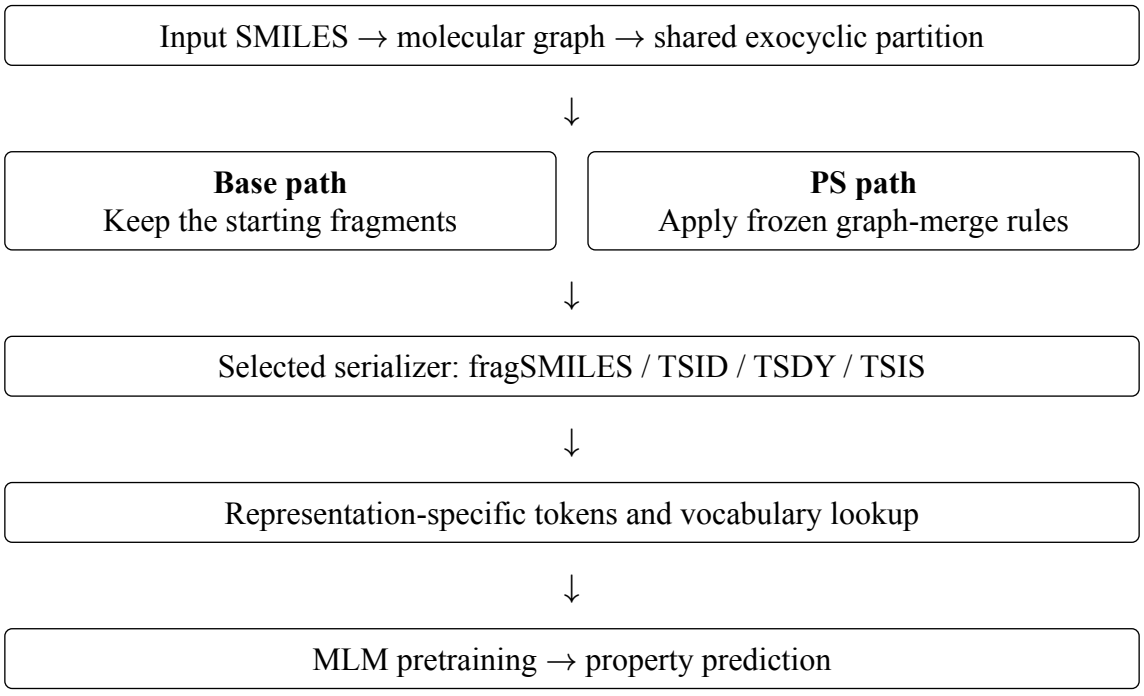


Figure 8: Shared-fragmentation pipeline. The base path retains the original fragments; the principal-subgraph (PS) path merges adjacent fragments before serialization.

RDKit parses molecules and constructs fragments (RDKit, n.d.). NetworkX identifies graph bridges and connected components (Hagberg et al., 2008). The fragSMILES adapter uses chemicalgof for assembly (Mastrolorito et al., 2025); the t-SMILES adapters implement the published tree traversals on the partition (Wu et al., 2024a; Wu et al., 2024b).

3.2 Shared fragmentation and principal-subgraph merging

3.2.1 Exocyclic starting partition

All four representations (fragSMILES, TSID, TSDY, TSIS) use the same starting fragments. RDKit parses a canonical isomeric SMILES for each molecule. The partitioner selects non-ring single bonds using chemicalgof's exocyclic cut rule (Mastrolorito et al., 2025), including its restriction on charged atom pairs. It cuts a selected bond only if removing that bond disconnects its endpoints. This bridge check keeps bonds within a fragment whenever another path connects the same atoms.

Each connected group of atoms left after cutting forms one fragment. The implementation records its atom indices and the cut bonds in the original molecule. The fragment graph is therefore a tree for each connected molecular component, even when a fragment itself contains rings. The partitioner checks that every atom belongs to exactly one fragment and that the recorded cuts coincide with the fragment boundaries.

3.2.2 Serialization of the common partition

Each serializer receives the same atom groups and cut bonds. The formats differ in how they express the connections:

- **fragSMILES** uses chemicalgof to write the fragments with external connection annotations.
- **TSID** marks both ends of each cut with matching labelled dummy atoms, such as `[1*]`. It writes the fragments in breadth-first order on a first-child/next-sibling binary tree.
- **TSDY** uses the same traversal and keeps the dummy atoms, but removes their pairing identifiers (i.e., `[1*]` becomes `[*]`).
- **TSIS** retains the pairing identifiers and traverses the original fragment tree in breadth-first order, without binary-tree placeholders.

Each tokenizer builds its own vocabulary from the resulting strings. This allows the four notations to be compared at the same fragment boundaries.

3.2.3 Learning and applying PS merges

The merger adapts the principal-subgraph mining procedure described by Kong et al. (Kong et al., 2022) by starting from exocyclic fragments instead of individual atoms. It learns an ordered set of rules from the tokenizer-training corpus:

1. For each pair of adjacent fragments, form their union and count occurrences of its canonical SMILES across the corpus. The matching key retains bond orders, charges, and isotopes but omits stereochemistry. The original graph retains the stereochemical information for serialization.
2. Select the most frequent union and merge its non-overlapping occurrences. Lexical order resolves frequency ties, and bond-index order resolves competing occurrences within a molecule.

3. Update counts only for molecules containing the selected union. Repeat until the rule budget is reached or no candidate meets the minimum frequency.

At encoding time, all four serializers use the same frozen rules in the same order. Each merge removes one boundary between neighbouring fragments, preserving the tree structure. The merge budget controls which fragments can form. The tokenizer vocabulary budget separately controls which serialized units receive their own token ID and which map to UNK.

3.3 Enhanced TSDY

3.3.1 Core words and attachment sites

In standard TSDY, dummy atoms mark attachment sites inside each fragment string. A benzene ring with one substituent and a benzene ring with two substituents therefore occupy different entries in a whole-fragment vocabulary. Enhanced TSDY uses the same exocyclic partition without PS merging. It caps the cut sites with hydrogen and writes each fragment as a canonical SMILES core. Both forms then use the ring word `c1ccccc1`. Separate tokens identify the attachment sites. This separation follows the clean-fragment approach of fragSMILES (Mastrolorito et al., 2025) and retains TSDY's tree traversal (Wu et al., 2024a).

3.3.2 Token stream and decoding

The encoder writes the cores in breadth-first order on a first-child/next-sibling binary tree. Each core string is one token. The decoder reads the tree to identify each fragment's parent. An internal empty slot requires a `<NULL>` token, while trailing empty slots can be omitted. A `<DOT>` starts a new disconnected component.

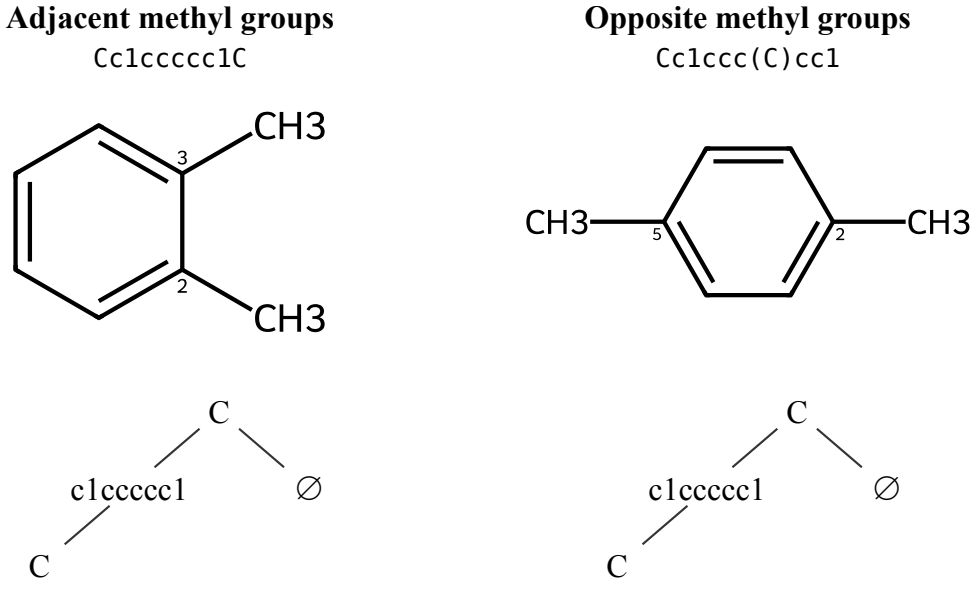


Figure 9: Enhanced-TSDY token streams for two dimethylbenzene isomers. Ring labels give core atom indices. Traversal proceeds from left to right at each level. The left branch points to the first child and the right branch to the next sibling. The empty sibling slot (∅) emits `<NULL>`.

For each child, the decoder restores a single bond between the parent and child cores and removes a hydrogen cap at each endpoint. If a core has only one hydrogen-bearing atom, the decoder can infer that endpoint. Otherwise, the encoder writes the atom's index in the parsed core after the child word (for example, `<N1>` denotes atom index 1, counting from zero). When both endpoints need indices, the parent's comes first.

In Figure 9, the first methyl group attaches to ring atom 2 in both molecules. The second methyl group attaches to atom 3 in the adjacent isomer and atom 5 in the opposite isomer. The methyl core has only one atom, so it needs no index.

Token entry	**Meaning and examples**
Core SMILES	One word per core. For ethanol, `CCO` becomes `C C <NULL> O`.
`<NULL>`, `<DOT>`	Empty tree slot (ethanol above) and component separator. The salt `[Na+].[Cl-]` becomes `[Cl-] <DOT> [Na+]`.
`<N0>` to `<N63>`	Small nonnegative integers for atom indices and stereo records.
`<BIG>`, digits, `</BIG>`	For indices of 64 or more, the markers enclose the digits of one integer, as in `<BIG> 6 4 </BIG>`. Reusing digit tokens keeps the vocabulary size fixed as indices grow.
`<ACW>`, `<ACCW>`	Tetrahedral stereo suffixes for lactic acid: `C[C@H](O)C(=O)O` ends in `<ACCW> <N1>`. `C[C@@H](O)C(=O)O` ends in `<ACW> <N1>`.
`<B>`	For `F/C=C/F`, the stereo suffix is `<B> <N1> <N0> <N1> <N1> <N3> <N0> <N0> <N2> <N0>`.

Table 1: Main token families in enhanced TSDY. Grammar entries are separate vocabulary words, alongside chemical-core words.

3.3.3 Stereochemistry and chemistry scope

Molecules can have identical atoms and bonds but differ in how their atoms are arranged in space. Enhanced TSDY writes this information after the tree of each component.

Tetrahedral centres. In lactic acid, the central carbon bonds to four different groups (H, OH, methyl, and carboxyl) pointing towards the corners of a tetrahedron[1]. These groups can occupy two mirror-image arrangements. The SMILES strings `C[C@H](O)C(=O)O` and `C[C@@H](O)C(=O)O` distinguish them. Both produce the same tree tokens, `C C <NULL> O <NULL> <NULL> C=O O`. The left panel of Figure 10 shows the first arrangement. The wedge marks the orientation recorded after the tree. The first stream appends `<ACCW> <N1>` and the second appends `<ACW> <N1>`. Here fragment 1 contains just the central carbon. The encoder orders its neighbours by fragment and atom index and records the orientation relative to that order. After reconnecting the bonds, the decoder uses the same ordering to recover the specified arrangement.

Double-bond stereochemistry. The `=` in `C=C` specifies a double bond (bond order two). The slash marks on its neighbouring bonds specify the relative arrangement of the attached

[1] See The RDKit Book, "Tetrahedral Stereogenic atoms".

groups[2]. In `F/C=C/F`, the fluorine atoms lie on opposite sides of the carbon-carbon double bond. In `F/C=C\F`, they lie on the same side, as drawn in the right panel of Figure 10. Both produce the tree tokens `F C=C <N0> <NULL> F <N1>`. The following record for the same-side form is grouped by field below. Each atom address lists its fragment index first and its atom index second.

Start	Carbon 1	Carbon 2	Relation	Fluorine 1	Fluorine 2
`<B>`	`<N1> <N0>`	`<N1> <N1>`	`<N2>`	`<N0> <N0>`	`<N2> <N0>`

Table 2: Double-bond stereo record for same-side difluoroethene.

Both carbons belong to fragment 1, the `C=C` core. The final two addresses identify the neighbours whose relative positions define the configuration. The opposite-side form uses `<N3>` in the relation field.

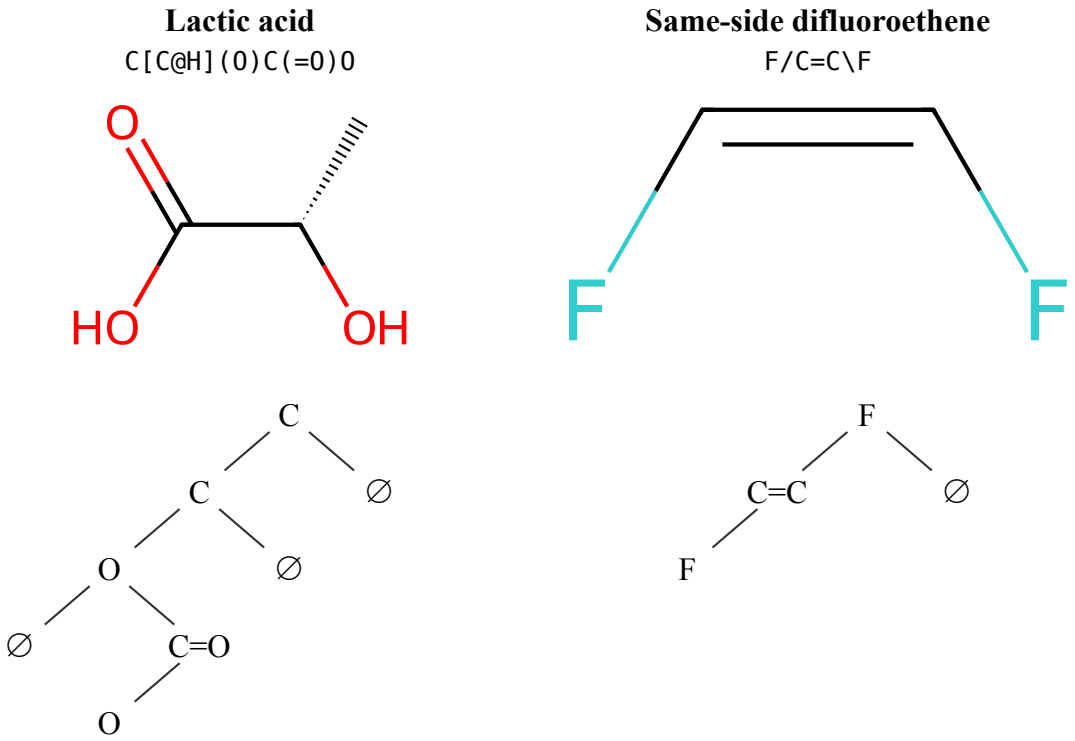


Figure 10: Enhanced-TSDY token streams for two molecules with stereochemistry.

Chemistry retained in cores. The word `[Na+]` preserves sodium's positive charge. The word `[13CH4]` preserves carbon-13, an isotope of carbon with a different neutron count from carbon-12. The methyl radical `[CH3]` retains its unpaired electron. Bonds inside a core also retain their type, as illustrated by the double bond in the `C=C` core above.

Scope. Enhanced TSDY excludes dative bonds. RDKit extends SMILES with arrows for dative bonds[3]. A minimal example is `N->[Fe+2]`. Nitrogen donates both bonding electrons to the iron ion, and the arrow points from donor to acceptor. A separate limit concerns the

[2]See The RDKit Book, "Stereogenic bonds".
[3]See The RDKit Book, "Dative bonds".

shape around an atom. In the lactic acid example, four groups point towards the corners of a tetrahedron. Around a square-planar platinum centre, four neighbours instead occupy the corners of a flat square[4]. For example, `Cl[Pt@SP1](Cl)(N)N` and `Cl[Pt@SP2](Cl)(N)N` specify different arrangements of the same neighbours. Enhanced TSDY has no token to preserve that distinction, so it rejects both inputs. Figure 11 shows the donor arrow and the two arrangements around platinum.

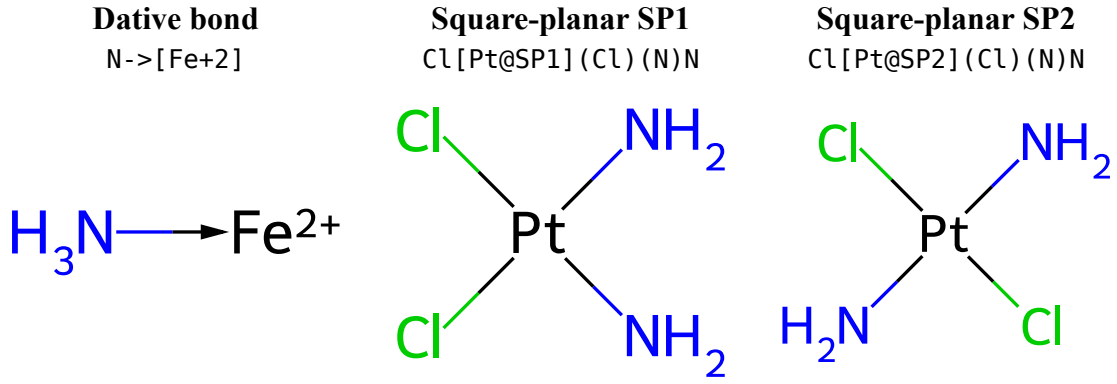


Figure 11: Dative bonding and square-planar stereochemistry outside the enhanced TSDY domain.

3.4 Reproducibility and implementation checks

Corpus separation. Across the eight shared-fragmentation configurations, the tokenizer-training rows had zero overlap with the MLM training, validation, and test rows. This check used PubChem row IDs. Distinct rows can still represent the same molecule. This separation prevents the tokenizer from learning its vocabulary from the exact records used for MLM training or evaluation.

Finetuning trials. Each trial used a different finetuning seed with the same tokenizer, pretrained model, and dataset split. Three trials were run for each configuration, and the results were averaged. Averaging reduces the influence of any single finetuning run on the comparison.

Vocabulary coverage. The intrinsic evaluation measures sequence length and unknown-token rates. Matching fragments by their atoms allows base and enhanced TSDY to be compared on the same fragment occurrences.

[4]See The RDKit Book, "Square planar".

4 Experimental methodology

4.1 Benchmark comparisons

Initial benchmark. The initial experiments compared character-, glyph-, atom-, and fragment-based tokenization on HIV and BBBP. Corresponding SMILES and SAFE configurations used the same tokenization method to test the effect of changing the input representation. For t-SMILES, each of TSSA, TSDY, TSID, and TSIS was evaluated with both BRICS and MMPA fragmentation. The main comparison used a target vocabulary of 5,000 entries for learned tokenizers. The APE baseline used the published vocabulary (Leon et al., 2024).

Shared fragmentation. Four representations (fragSMILES, TSDY, TSID, and TSIS) used the same exocyclic partition. Each was tested with and without principal-subgraph (PS) merging, giving eight conditions. Comparing the two conditions within a representation measures the effect of enabling merges. Comparing representations within either condition holds the atom partition fixed.

Enhanced TSDY. The comparison evaluates base and enhanced TSDY on HIV and BBBP from MoleculeNet (Wu et al., 2018). Both use the same exocyclic partition without merging, so the fragments are held fixed while their encoding changes.

4.2 Vocabulary construction and pretraining

Corpus selection. The controlled experiments used canonical SMILES from PubChem10M[5]. A fixed sample of 2,000,000 rows (seed 42) supplied the tokenizer corpus. The MLM sampler excluded these rows and selected 1,000,000 training, 100,000 validation, and 100,000 test rows. All controlled representations used these selections.

Vocabulary budget. The shared and enhanced experiments each allocated 5,000 entries per tokenizer, including special tokens. The PS miner used the tokenizer corpus, a budget of 1,000 merge rules, and a minimum frequency of 10. The same learned rules were applied to all four PS conditions. For enhanced TSDY, 81 grammar tokens and seven special tokens occupy fixed entries. The remaining 4,912 entries hold the most frequent chemical cores in the tokenizer corpus.

Model training. Each controlled tokenizer had its own RoBERTa encoder (Liu et al., 2019) pretrained from scratch with seed 42. The encoder had six layers, 12 attention heads, hidden size 768, and feed-forward size 1,536. Training used the Hugging Face Transformers Trainer (Wolf et al., 2020) with the settings in Table 3. Dynamic masking selected 15% of eligible positions as prediction targets. Tokenizer construction ran on CPU nodes (AMD EPYC 7763), while model training ran on GPU nodes (NVIDIA H100 96GB) on the NUS School of Computing cluster using Slurm (Appendix B).

[5]Dataset: PubChem10M SMILES/SELFIES.

Setting	MLM	Finetuning
Epochs	20	Up to 5 (early stopping)
Batch size per device	64	64
Learning rate	5×10^{-5}	5×10^{-5}
Optimizer	AdamW	AdamW
Weight decay	0.01	0.01
Learning-rate schedule	Linear	Linear
Warmup fraction	0.1	0.1

Table 3: Training settings for the controlled comparisons.

Model inputs allowed 510 content tokens plus two boundary tokens. Longer sequences were truncated for model training and evaluation. Vocabulary fitting and intrinsic length measurements used the complete sequence.

4.3 Downstream evaluation

Data splits. The binary labels indicate whether a compound inhibits HIV replication or crosses the blood-brain barrier, respectively (Wu et al., 2018). HIV and BBBP used DeepChem's scaffold splitter with target proportions of 80/10/10 (Ramsundar et al., 2019). This assigns molecules with the same scaffold to the same split. Keeping scaffold groups intact gives the counts in Table 4.

Task and protocol	Train	Validation	Test	Total
HIV	32,901	4,113	4,106	41,120
BBBP	1,640	205	194	2,039

Table 4: Number of molecules in each downstream data split.

Model selection and scoring. Each classification run started from the final MLM weights for its representation. The area under the receiver operating characteristic curve (ROC-AUC) measures how often the model ranks a positive example above a negative one, counting ties as half. Validation ROC-AUC was evaluated each epoch, with early stopping after three evaluations without improvement. The checkpoint with the highest validation ROC-AUC supplied the test predictions.

The downstream protocol uses three finetuning seeds (42, 43, and 44) per configuration. The split and pretrained encoder are fixed across trials. Each comparison reports the mean and sample standard deviation of the three test ROC-AUC values.

4.4 Intrinsic evaluation

The intrinsic comparison in Appendix D covers 33 tokenizer configurations. Each was evaluated on the same 53,159 input rows: 10,000 PubChem rows sampled from a saved test corpus (seed 42), 41,120 HIV rows, and 2,039 BBBP rows. The audit used each tokenizer's saved vocabulary and recorded metrics separately for PubChem and for each HIV and BBBP split. Appendix D presents all 33 configurations for each input group. Section 5.4 reports selected comparisons on the downstream test splits.

4.4.1 Encoding and vocabulary metrics

- **Encoding success rate.** Fraction of attempted rows that produced a token sequence successfully. The strict SAFE audit counted conversion failures even when the historical training pipeline would have used a SMILES fallback.
- **Mean sequence length.** Mean number of content tokens per successfully encoded molecule, measured before truncation. Connection and tree tokens count towards length. Model boundary tokens and padding are excluded.
- **Token fertility (tokens per heavy atom).** Total content tokens divided by total non-hydrogen atoms across the successfully encoded molecules. This measures sequence length relative to molecular size.
- **UNK token rate.** Number of positions assigned the unknown-token ID divided by the number of content tokens across all successfully encoded molecules.
- **Molecules containing UNK rate.** Fraction of successfully encoded molecules with at least one unknown token. Each affected molecule counts once, regardless of how many unknown tokens it contains.

Paired comparisons used rows that both tokenizers encoded successfully.

4.4.2 Fragment sharing

- **Decorated forms per core.** For base and enhanced TSDY, fragments were paired by their atoms in the shared partition. For each clean core, the audit counted the distinct base TSDY fragment strings that mapped to it.
- **Fragment UNK rate.** Unknown fragment occurrences divided by all matched fragment occurrences. Excluding tree and attachment tokens gives both representations the same denominator.

5 Results and discussion

5.1 Initial benchmark of existing tokenizers

String baselines. Plain SMIRK on SMILES had the highest observed mean test ROC-AUC among the 25 initial configurations on both HIV (0.7755 ± 0.0120) and BBBP (0.7467 ± 0.0082). Adding structurally pre-split glyph-pair encoding (SMIRK-SGPE) gave a nearly identical HIV mean (0.7752 ± 0.0199). Two comparisons using ordinary SMILES BPE as the baseline separate changes to the tokenizer from changes to the input representation.

- **Change the tokenization rule.** Adding structural pre-splitting[6] increased mean test ROC-AUC slightly on HIV (+0.0026) and more on BBBP (+0.0222).
- **Change the molecular representation.** Keeping the BPE training rule fixed, switching from SMILES to SAFE lowered the mean on both HIV (-0.0042) and BBBP (-0.0423). SAFE had a lower BBBP mean in all eight paired comparisons. On HIV, APE was the only tokenizer for which SAFE had a higher mean (+0.0227).

Appendix C reports the full three-seed results.

Fragmentation and assembly. Switching the fragmentation method from BRICS to MMPA increased mean ROC-AUC for every t-SMILES notation on both tasks (Table 5). The effect of assembly notation varied by task. With BRICS, TSSA and TSDY both ranked above TSID and TSIS on both tasks. With MMPA, TSIS had the highest HIV mean (0.7268), while TSDY had the highest BBBP mean (0.6954).

TSID and TSIS include numbered attachment sites in fragment words (Wu et al., 2024a; Wu et al., 2024b). Changing a connection label from `[1*]` to `[2*]` can split occurrences of the same motif across vocabulary entries. TSSA and TSDY avoid that source of duplication. This provides a reason to test vocabulary coverage directly. The reversal on MMPA HIV shows why the identifiers alone cannot explain the predictive ranking.

Notation	HIV (BRICS)	HIV (MMPA)	BBBP (BRICS)	BBBP (MMPA)
TSSA	0.6015 ± 0.0162	0.7015 ± 0.0189	0.6175 ± 0.0325	0.6776 ± 0.0042
TSDY	0.6147 ± 0.0164	0.7007 ± 0.0248	0.6259 ± 0.0054	0.6954 ± 0.0058
TSID	0.5766 ± 0.0190	0.7054 ± 0.0337	0.5538 ± 0.0281	0.6874 ± 0.0047
TSIS	0.5586 ± 0.0081	0.7268 ± 0.0212	0.5603 ± 0.0339	0.6569 ± 0.0102

Table 5: Initial test ROC-AUC by fragmentation rule and t-SMILES notation. Values are means $\pm$ sample standard deviations across finetuning seeds 42, 43, and 44.

Comparison with fragSMILES. The shared base fragSMILES reference from Table 6 scored 0.7615 on HIV and 0.7184 on BBBP, above all the t-SMILES variants in Table 5. Two choices changed together: fragSMILES cut exocyclic single bonds and recorded connections outside its fragment words. The scores alone cannot distinguish their contributions because they do not use the same fragmentation rule (BRICS/MMPA vs. cutting

[6] During vocabulary training, structural pre-splitting isolates branch parentheses, ring numbers, bracketed atoms, component separators, and directional bond marks. For example, `CC(=O)O` yields the spans `CC`, `(`, `=O`, `)`, and `O`. BPE learns merges within these spans, while PCATT uses their counts to learn vocabulary entries.

exocyclic single bonds). Section 5.2 compares serializers on the same fragmentation rule and measures the effect of merging the fragments. Section 5.3 keeps the TSDY partition fixed and changes how attachment sites are encoded.

5.2 Shared fragmentation and PS merging

With the base exocyclic partition, TSID had the highest mean HIV score and fragSMILES had the highest mean BBBP score (Table 6). PS merging reduced the mean BBBP score for every representation. On HIV, TSDY was the only representation whose mean increased after merging.

Representation	HIV ROC-AUC	BBBP ROC-AUC
fragSMILES	0.7615 ± 0.0130	0.7184 ± 0.0102
PS-fragSMILES	0.7310 ± 0.0112	0.6911 ± 0.0086
TSID	0.7632 ± 0.0074	0.7004 ± 0.0026
PS-TSID	0.7342 ± 0.0129	0.6485 ± 0.0214
TSDY	0.7181 ± 0.0306	0.6909 ± 0.0123
PS-TSDY	0.7334 ± 0.0264	0.6827 ± 0.0105
TSIS	0.7436 ± 0.0139	0.7089 ± 0.0051
PS-TSIS	0.7188 ± 0.0177	0.6436 ± 0.0188

Table 6: Test ROC-AUC with the shared exocyclic partition, before and after PS merging. Values are means $\pm$ sample standard deviations across three finetuning seeds.

The effect of merging also depended on the assembly notation. On BBBP, PS merging lowered mean ROC-AUC by 5.19 percentage points for TSID and 6.53 for TSIS, compared with 0.82 for TSDY. On HIV, the TSID and TSIS means also fell, while TSDY rose from 0.7181 to 0.7334. TSDY was less adversely affected by merging than TSID or TSIS on either task. The PubChem measurements in Table 8 support the coverage part of this explanation. After merging, the UNK token rate was 4.97% for TSDY, compared with 11.43% for TSID and 16.32% for TSIS. These rates describe vocabulary coverage. They do not isolate its effect on ROC-AUC.

5.3 Enhanced TSDY on HIV and BBBP

Enhanced TSDY scored higher than base TSDY at each matched seed on both tasks. The mean gains were 1.91 ROC-AUC percentage points on HIV and 3.03 on BBBP (Table 7).

Representation	HIV ROC-AUC	BBBP ROC-AUC
Base TSDY[7]	0.7363 ± 0.0104	0.6909 ± 0.0123
Enhanced TSDY	0.7554 ± 0.0120	0.7212 ± 0.0060

Table 7: Base and enhanced TSDY test ROC-AUC on HIV and BBBP. Values are means $\pm$ sample standard deviation. Both representations use the same molecules within each split.

HIV training inputs. Of the 32,901 HIV training rows, only source row 35278 could not be encoded by enhanced TSDY. RDKit parses two sulfur-to-iron bonds in this molecule as

[7]The HIV baseline was finetuned again on the same training rows as enhanced TSDY. The BBBP baseline is reused from Table 6.

dative bonds, which the encoder does not support. Figure 12 compares the dataset string with the canonical SMILES produced by RDKit.

Dataset SMILES

```
C1C[N+]23CCC[N+]4(C1)CC[S-]1[Fe+2]24([SH+]CC3)
[S-]2CC[N+]34CCC[N+]5(CCC3)CC[SH+][Fe+2]1254
```

RDKit canonical SMILES

```
C1C[N+]23CCC[N+]4(C1)CC[S-]1->[Fe+2]567[SH+]CC
[N+]58CCC[N+]6(CCC8)CC[S-]7->[Fe+2]124[SH+]CC3
```

Figure 12: HIV training row 35278. Each `->` in the canonical string marks a dative bond.

Both TSDY models therefore used the same 32,900 training rows. All 4,113 validation rows and all 4,106 test rows passed the encoding check and were retained.

5.4 Sequence length and vocabulary coverage

On PubChem, structural pre-splitting increased the mean length of SMILES BPE sequences from 7.18 to 28.27 tokens, and its mean prediction scores were higher on both tasks in Section 5.1. For TSDY, switching from BRICS to MMPA also lengthened the sequences, from 9.01 to 21.08 tokens per molecule, but reduced the UNK token rate from 10.19% to 1.57%.

Effect of PS merging. Merging reduced token fertility for every shared representation, but increased the fraction of content tokens mapped to UNK (Table 8). TSID and TSDY illustrate why both measures matter. Their fertility fell from 1.02 to 0.46 in both cases, yet the merged TSID vocabulary left 11.43% of token positions unknown compared with 4.97% for TSDY. The coverage difference is consistent with the cost of storing numbered attachment sites discussed in Section 5.1.

Representation	Fertility	UNK token rate (%)
fragSMILES	$0.79 \rightarrow 0.46$	$0.25 \rightarrow 0.48$
TSID	$1.02 \rightarrow 0.46$	$3.11 \rightarrow 11.43$
TSDY	$1.02 \rightarrow 0.46$	$0.72 \rightarrow 4.97$
TSIS	$0.82 \rightarrow 0.32$	$3.83 \rightarrow 16.32$

Table 8: Token fertility and UNK token rate on 10,000 PubChem molecules. Each arrow runs from the base value to the PS value. Fertility counts content tokens per heavy atom. The UNK token rate uses all content-token positions.

Sharing across attachment patterns. The 10,000 PubChem molecules contained 108,850 fragment occurrences. Counting each distinct fragment spelling once gave 4,113 decorated forms in base TSDY and 1,815 clean cores in enhanced TSDY. For example, pyridine occurred 906 times across 40 decorated forms, all of which used the same enhanced core word, `c1ccncc1`. Table 9 compares vocabulary coverage occurrence by occurrence, pairing the two encodings of the same group of atoms.

Input set	Fragment occurrences	Base TSDY fragment UNK rate (%)	Enhanced TSDY fragment UNK rate (%)
PubChem	108,850	1.71	0.46
HIV test	31,240	10.58	6.78
BBBP test	1,873	10.20	2.08

Table 9: Fragment UNK rates for matched base and enhanced TSDY fragments. Rates count repeated occurrences and exclude tree and attachment tokens.

On BBBP, the fragment UNK rate fell from 10.20% to 2.08%, even as mean sequence length rose from 23.44 to 28.30 tokens. The same core word can cover several attachment patterns, while the tokens that specify those attachments still occupy sequence positions.

5.5 Discussion

The initial benchmark showed a consistent benefit from MMPA over BRICS, while the ranking of assembly notations depended on the task. At a vocabulary size of 5,000, merging neighbouring fragments saved sequence positions but made the resulting fragment words harder to cover. Every PS variant had a lower mean BBBP score than its base counterpart, and the HIV results were mixed. Separating attachment information gave a different outcome. Enhanced TSDY reused core words across attachment patterns and had a higher mean ROC-AUC than base TSDY on both tasks, including BBBP, where the sequences became longer. These results favour core sharing for the TSDY comparison and show why token fertility alone is insufficient for selecting a molecular tokenizer.

6 Conclusions

6.1 Summary of findings

The initial benchmark identified SMIRK as a strong glyph-based baseline. Among the t-SMILES tokenizers, changing the fragmentation rule from BRICS to MMPA improved prediction across all tested notations, while the best assembly notation depended on the task.

The controlled experiments showed that shortening a sequence and improving vocabulary coverage are distinct objectives. Principal-subgraph merging shortened sequences, but more fragments fell outside the vocabulary and prediction did not improve consistently. Enhanced TSDY instead used a common core word for fragments with different attachment patterns. This improved fragment coverage and prediction over base TSDY on both tasks, even when the attachment records made sequences longer.

6.2 Limitations

Task scope. The downstream evaluation covered HIV and BBBP classification with one scaffold split per dataset. It leaves open how the tokenizers perform on other molecular properties or in molecule generation.

Attachment encoding. Enhanced TSDY changed the core words, connection syntax, and stereo records together. Their individual contributions to prediction remain uncertain.

Chemical coverage. Enhanced TSDY supports a limited set of bond types and stereochemical arrangements. Rare cores can also fall outside the learned vocabulary.

6.3 Recommendations for future work

Isolate core sharing. Compare two tokenizers with identical fragment boundaries, traversal, and connection records. At the same vocabulary size, give one tokenizer a shared word per core and the other distinct words for different attachment patterns.

Vary the merge count. Test whether fewer PS merges can shorten sequences while keeping more fragments in the vocabulary.

Broaden property prediction. Evaluate additional datasets, including a regression task such as predicting molecular solubility. This would test whether the findings extend beyond binary classification.

Test molecule generation. Train models to generate token sequences and compare how often those sequences decode into valid molecules. This would test whether the connection syntax is practical for producing new structures.

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Appendix A: Declaration of AI assistance

I used OpenAI Codex to discuss implementation designs and plan experiments. It helped generate and revise code, analyze experimental results, as well as draft and format this report. I take responsibility for the submitted code, the report's accuracy, and the interpretation of the results.

Appendix B: Reproducibility details

Computing environment. Tokenizer construction and principal-subgraph mining used AMD EPYC 7763 CPU nodes. Training jobs used NVIDIA H100 96GB GPU nodes. The training environment used PyTorch 2.10.0, CUDA 12.8, RDKit 2025.9.4 and Hugging Face Transformers 4.57.6.

Code. The repository contains the source code for the project.
https://github.com/ncduy0303/molecule-tokenization

Data. The tokenizer and pretraining corpora use this PubChem10M dataset.
https://huggingface.co/datasets/mikemayuare/PubChem10M_SMILES_SELFIES

For downstream finetuning, the loader downloads the MoleculeNet HIV and BBBP datasets from DeepChem as CSV files.

- HIV: https://deepchemdata.s3-us-west-1.amazonaws.com/datasets/HIV.csv
- BBBP: https://deepchemdata.s3-us-west-1.amazonaws.com/datasets/BBBP.csv

Experiment records. The Weights & Biases project (https://wandb.ai/ncduy0303/molecule-tokenization) provides two saved views:

- FYP Final Report groups the 207 downstream runs into 69 comparisons with three finetuning seeds each.
- FYP Pretraining lists 30 pretraining MLM runs.

Appendix C: Downstream results by configuration

Scores are mean test ROC-AUC ± sample standard deviation over finetuning seeds 42, 43, and 44. The table lists the 25 initial configurations and the shared base fragSMILES reference from Table 6. SBPE, SPCATT, and SGPE denote structurally pre-split BPE, PCATT, and glyph-pair encoding, respectively.

Family	Configuration	HIV ROC-AUC	BBBP ROC-AUC
Character	SMILES BPE	0.7529 ± 0.0190	0.7136 ± 0.0066
	SMILES SBPE	0.7554 ± 0.0253	0.7358 ± 0.0098
	SMILES SPCATT	0.7688 ± 0.0141	0.7287 ± 0.0028
	SAFE BPE	0.7487 ± 0.0171	0.6713 ± 0.0045
	SAFE SBPE	0.7506 ± 0.0200	0.6725 ± 0.0289
	SAFE SPCATT	0.7413 ± 0.0090	0.6608 ± 0.0317
Glyph	SMILES SMIRK	0.7755 ± 0.0120	0.7467 ± 0.0082
	SMILES SMIRK-SGPE	0.7752 ± 0.0199	0.7416 ± 0.0045
	SMILES SMIRK-SPCATT	0.7672 ± 0.0374	0.7237 ± 0.0129
	SAFE SMIRK	0.7362 ± 0.0169	0.6692 ± 0.0171
	SAFE SMIRK-GPE	0.7440 ± 0.0058	0.6943 ± 0.0105
	SAFE SMIRK-SGPE	0.7379 ± 0.0238	0.6747 ± 0.0417
	SAFE SMIRK-SPCATT	0.7239 ± 0.0039	0.7172 ± 0.0021
Atom	SMILES APE	0.7409 ± 0.0300	0.7085 ± 0.0095
	SMILES SPE	0.7678 ± 0.0128	0.7219 ± 0.0026
	SAFE APE	0.7636 ± 0.0037	0.6728 ± 0.0174
	SAFE SPE	0.7197 ± 0.0194	0.6837 ± 0.0181
Fragment	TSSA BRICS	0.6015 ± 0.0162	0.6175 ± 0.0325
	TSSA MMPA	0.7015 ± 0.0189	0.6776 ± 0.0042
	TSDY BRICS	0.6147 ± 0.0164	0.6259 ± 0.0054
	TSDY MMPA	0.7007 ± 0.0248	0.6954 ± 0.0058
	TSID BRICS	0.5766 ± 0.0190	0.5538 ± 0.0281
	TSID MMPA	0.7054 ± 0.0337	0.6874 ± 0.0047
	TSIS BRICS	0.5586 ± 0.0081	0.5603 ± 0.0339
	TSIS MMPA	0.7268 ± 0.0212	0.6569 ± 0.0102
	Base fragSMILES	0.7615 ± 0.0130	0.7184 ± 0.0102

Table 10: Initial benchmark test ROC-AUC by tokenizer family.

Tokenizer	HIV ΔROC-AUC	BBBP ΔROC-AUC
BPE	−0.0042	−0.0423
SBPE	−0.0048	−0.0632
SPCATT	−0.0274	−0.0679
SMIRK	−0.0393	−0.0775
SMIRK-SGPE	−0.0373	−0.0669
SMIRK-SPCATT	−0.0433	−0.0065
APE	+0.0227	−0.0357
SPE	−0.0481	−0.0382

Table 11: Initial mean test ROC-AUC differences (SAFE minus SMILES).

Appendix D: Intrinsic results by configuration

Each pair of tables covers one input group. Length, fertility, and UNK measures use the successfully encoded rows.

PubChem

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
SAFE APE	9,717	24.28	0.95	19.56	100.00
SAFE BPE	9,717	11.18	0.44	0.00	0.00
SAFE SBPE	9,717	40.77	1.60	0.00	0.00
SAFE SMIRK	9,717	56.29	2.21	0.00	0.01
SAFE SMIRK GPE	9,717	11.13	0.44	0.00	0.00
SAFE SMIRK SGPE	9,717	40.77	1.60	0.00	0.00
SAFE SMIRK SPCATT	9,717	40.77	1.60	0.00	0.00
SAFE SPCATT	9,717	40.77	1.60	0.00	0.00
SAFE SPE	9,717	10.69	0.42	0.00	0.00
SMILES APE	10,000	8.95	0.36	0.87	6.71
SMILES BPE	10,000	7.18	0.29	0.00	0.00
SMILES SBPE	10,000	28.27	1.12	0.00	0.00
SMILES SMIRK	10,000	44.23	1.76	0.00	0.01
SMILES SMIRK SPCATT	10,000	39.81	1.58	0.00	0.00
SMILES SPCATT	10,000	28.27	1.12	0.00	0.00
SMILES SPE	10,000	7.12	0.28	0.00	0.00

Table 12: SMILES and SAFE tokenizers on the PubChem inputs (10,000 attempted rows).

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
TSDY BRICS	10,000	9.01	0.36	10.19	79.67
TSDY MMPA	10,000	21.08	0.84	1.57	30.54
TSID BRICS	10,000	9.01	0.36	14.02	92.38
TSID MMPA	10,000	21.08	0.84	4.57	68.67
TSIS BRICS	10,000	5.83	0.23	21.66	92.38
TSIS MMPA	10,000	16.72	0.66	5.76	68.66
TSSA BRICS	10,000	13.57	0.54	5.97	72.75
TSSA MMPA	10,000	28.58	1.13	0.50	13.94
Enhanced TSDY	10,000	24.99	0.99	0.20	4.93
Base TSDY	10,000	25.62	1.02	0.72	17.61
Base TSID	10,000	25.62	1.02	3.11	59.00
Base TSIS	10,000	20.77	0.82	3.83	59.00
Base fragSMILES	10,000	19.88	0.79	0.25	4.92
PS-TSDY	10,000	11.51	0.46	4.97	48.38
PS-TSID	10,000	11.51	0.46	11.43	79.48
PS-TSIS	10,000	8.06	0.32	16.32	79.48
PS-fragSMILES	10,000	11.65	0.46	0.48	5.54

Table 13: Fragment tokenizers on the PubChem inputs (10,000 attempted rows).

HIV training

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
SAFE APE	27,906	23.00	0.87	17.98	100.00
SAFE BPE	27,906	12.58	0.47	0.00	0.00
SAFE SBPE	27,906	40.97	1.55	0.00	0.00
SAFE SMIRK	27,906	56.40	2.13	0.00	0.00
SAFE SMIRK GPE	27,906	12.52	0.47	0.00	0.00
SAFE SMIRK SGPE	27,906	40.97	1.55	0.00	0.00
SAFE SMIRK SPCATT	27,906	40.99	1.55	0.00	0.00
SAFE SPCATT	27,906	41.00	1.55	0.00	0.00
SAFE SPE	27,906	12.08	0.46	0.24	2.67
SMILES APE	32,901	10.08	0.40	3.41	15.65
SMILES BPE	32,901	8.33	0.33	0.00	0.00
SMILES SBPE	32,901	29.21	1.16	0.00	0.00
SMILES SMIRK	32,901	44.56	1.76	0.00	0.00
SMILES SMIRK SPCATT	32,901	40.10	1.59	0.00	0.00
SMILES SPCATT	32,901	29.24	1.16	0.00	0.00
SMILES SPE	32,901	8.08	0.32	1.41	8.30

Table 14: SMILES and SAFE tokenizers on the HIV training inputs (32,901 attempted rows).

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
TSDY BRICS	32,896	8.24	0.33	14.63	95.04
TSDY MMPA	32,896	18.19	0.72	4.98	72.72
TSID BRICS	32,896	8.24	0.33	17.94	98.46
TSID MMPA	32,896	18.19	0.72	8.06	92.63
TSIS BRICS	32,896	4.62	0.18	29.97	98.46
TSIS MMPA	32,896	13.66	0.54	10.37	92.63
TSSA BRICS	32,896	11.56	0.46	9.55	90.73
TSSA MMPA	32,896	24.69	0.98	2.64	53.39
Enhanced TSDY	32,900	23.10	0.91	1.21	26.09
Base TSDY	32,901	24.15	0.96	2.70	51.79
Base TSID	32,901	24.15	0.96	5.00	80.33
Base TSIS	32,901	19.10	0.76	6.32	80.33
Base fragSMILES	32,901	19.01	0.75	1.89	30.07
PS-TSDY	32,901	12.44	0.49	8.28	73.86
PS-TSID	32,901	12.44	0.49	14.59	91.88
PS-TSIS	32,901	8.73	0.35	20.78	91.88
PS-fragSMILES	32,901	12.98	0.51	2.89	31.47

Table 15: Fragment tokenizers on the HIV training inputs (32,901 attempted rows).

HIV validation

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
SAFE APE	3,197	24.76	0.85	17.05	100.00
SAFE BPE	3,197	14.35	0.49	0.00	0.00
SAFE SBPE	3,197	43.51	1.49	0.00	0.00
SAFE SMIRK	3,197	62.07	2.12	0.00	0.06
SAFE SMIRK GPE	3,197	14.28	0.49	0.00	0.00
SAFE SMIRK SGPE	3,197	43.51	1.49	0.00	0.00
SAFE SMIRK SPCATT	3,197	43.56	1.49	0.00	0.00
SAFE SPCATT	3,197	43.58	1.49	0.00	0.00
SAFE SPE	3,197	13.62	0.47	0.42	5.32
SMILES APE	4,113	12.36	0.45	5.60	17.75
SMILES BPE	4,113	10.72	0.39	0.00	0.00
SMILES SBPE	4,113	32.38	1.17	0.00	0.00
SMILES SMIRK	4,113	50.35	1.81	0.00	0.05
SMILES SMIRK SPCATT	4,113	45.08	1.62	0.00	0.00
SMILES SPCATT	4,113	32.44	1.17	0.00	0.00
SMILES SPE	4,113	9.96	0.36	1.79	12.25

Table 16: SMILES and SAFE tokenizers on the HIV validation inputs (4,113 attempted rows).

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
TSDY BRICS	4,110	8.34	0.30	16.72	97.79
TSDY MMPA	4,110	16.97	0.61	7.43	87.47
TSID BRICS	4,110	8.34	0.30	19.56	99.17
TSID MMPA	4,110	16.97	0.61	10.24	97.69
TSIS BRICS	4,110	4.13	0.15	33.55	99.17
TSIS MMPA	4,110	11.92	0.43	13.36	97.71
TSSA BRICS	4,110	11.22	0.40	11.77	95.79
TSSA MMPA	4,110	23.00	0.83	4.69	74.06
Enhanced TSDY	4,113	21.03	0.76	2.99	53.93
Base TSDY	4,113	21.49	0.77	4.62	74.35
Base TSID	4,113	21.49	0.77	7.01	93.00
Base TSIS	4,113	16.65	0.60	9.05	93.00
Base fragSMILES	4,113	17.77	0.64	4.01	56.36
PS-TSDY	4,113	12.79	0.46	10.03	86.68
PS-TSID	4,113	12.79	0.46	15.40	96.94
PS-TSIS	4,113	9.01	0.32	21.85	96.94
PS-fragSMILES	4,113	13.58	0.49	5.38	57.79

Table 17: Fragment tokenizers on the HIV validation inputs (4,113 attempted rows).

HIV test

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
SAFE APE	2,901	22.58	0.81	16.19	100.00
SAFE BPE	2,901	13.23	0.47	0.00	0.00
SAFE SBPE	2,901	41.08	1.47	0.00	0.00
SAFE SMIRK	2,901	57.41	2.05	0.00	0.00
SAFE SMIRK GPE	2,901	13.14	0.47	0.00	0.00
SAFE SMIRK SGPE	2,901	41.08	1.47	0.00	0.00
SAFE SMIRK SPCATT	2,901	41.09	1.47	0.00	0.00
SAFE SPCATT	2,901	41.09	1.47	0.00	0.00
SAFE SPE	2,901	12.70	0.45	0.08	1.00
SMILES APE	4,106	10.65	0.42	2.77	10.20
SMILES BPE	4,106	9.28	0.37	0.00	0.00
SMILES SBPE	4,106	29.04	1.15	0.00	0.00
SMILES SMIRK	4,106	44.44	1.76	0.00	0.00
SMILES SMIRK SPCATT	4,106	39.84	1.58	0.00	0.00
SMILES SPCATT	4,106	29.06	1.15	0.00	0.00
SMILES SPE	4,106	8.97	0.36	0.79	6.41

Table 18: SMILES and SAFE tokenizers on the HIV test inputs (4,106 attempted rows).

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
TSDY BRICS	4,106	7.16	0.28	16.77	96.57
TSDY MMPA	4,106	14.60	0.58	7.42	88.72
TSID BRICS	4,106	7.16	0.28	19.74	99.00
TSID MMPA	4,106	14.60	0.58	10.18	97.91
TSIS BRICS	4,106	3.63	0.14	36.49	99.00
TSIS MMPA	4,106	10.41	0.41	13.86	97.91
TSSA BRICS	4,106	9.65	0.38	11.71	94.25
TSSA MMPA	4,106	19.34	0.77	4.68	76.38
Enhanced TSDY	4,106	18.13	0.72	2.84	50.00
Base TSDY	4,106	18.66	0.74	4.57	74.40
Base TSID	4,106	18.66	0.74	6.81	92.26
Base TSIS	4,106	14.22	0.56	8.94	92.26
Base fragSMILES	4,106	15.57	0.62	3.60	52.73
PS-TSDY	4,106	11.80	0.47	9.53	86.53
PS-TSID	4,106	11.80	0.47	14.55	96.25
PS-TSIS	4,106	8.24	0.33	20.85	96.25
PS-fragSMILES	4,106	12.25	0.48	4.71	54.04

Table 19: Fragment tokenizers on the HIV test inputs (4,106 attempted rows).

BBBP training

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
SAFE APE	1,428	19.50	0.82	18.63	100.00
SAFE BPE	1,428	10.97	0.46	0.00	0.00
SAFE SBPE	1,428	36.51	1.53	0.00	0.00
SAFE SMIRK	1,428	47.88	2.01	0.00	0.00
SAFE SMIRK GPE	1,428	10.97	0.46	0.00	0.00
SAFE SMIRK SGPE	1,428	36.51	1.53	0.00	0.00
SAFE SMIRK SPCATT	1,428	36.56	1.53	0.00	0.00
SAFE SPCATT	1,428	36.57	1.54	0.00	0.00
SAFE SPE	1,428	10.62	0.45	0.53	3.08
SMILES APE	1,640	16.14	0.71	36.42	41.34
SMILES BPE	1,640	14.08	0.62	0.00	0.00
SMILES SBPE	1,640	30.31	1.33	0.00	0.00
SMILES SMIRK	1,640	43.72	1.92	0.00	0.00
SMILES SMIRK SPCATT	1,640	40.48	1.78	0.00	0.00
SMILES SPCATT	1,640	30.37	1.33	0.00	0.00
SMILES SPE	1,640	9.98	0.44	15.37	34.33

Table 20: SMILES and SAFE tokenizers on the BBBP training inputs (1,640 attempted rows).

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
TSDY BRICS	1,639	8.27	0.36	13.85	90.18
TSDY MMPA	1,639	18.40	0.81	4.59	67.05
TSID BRICS	1,639	8.27	0.36	16.40	94.94
TSID MMPA	1,639	18.40	0.81	7.07	85.72
TSIS BRICS	1,639	4.84	0.21	26.72	94.94
TSIS MMPA	1,639	14.08	0.62	8.96	85.72
TSSA BRICS	1,639	11.75	0.52	8.62	83.34
TSSA MMPA	1,639	24.12	1.06	2.00	41.55
Enhanced TSDY	1,640	24.76	1.09	0.82	17.26
Base TSDY	1,640	21.62	0.95	3.84	59.21
Base TSID	1,640	21.62	0.95	5.16	77.07
Base TSIS	1,640	16.99	0.75	6.57	77.07
Base fragSMILES	1,640	17.20	0.76	8.02	39.70
PS-TSDY	1,640	12.17	0.53	9.23	78.66
PS-TSID	1,640	12.17	0.53	13.18	90.12
PS-TSIS	1,640	8.57	0.38	18.72	90.12
PS-fragSMILES	1,640	12.87	0.57	10.72	40.79

Table 21: Fragment tokenizers on the BBBP training inputs (1,640 attempted rows).

BBBP validation

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
SAFE APE	188	28.98	0.88	17.31	100.00
SAFE BPE	188	17.28	0.52	0.00	0.00
SAFE SBPE	188	54.98	1.67	0.00	0.00
SAFE SMIRK	188	71.89	2.18	0.00	0.00
SAFE SMIRK GPE	188	17.28	0.52	0.00	0.00
SAFE SMIRK SGPE	188	54.98	1.67	0.00	0.00
SAFE SMIRK SPCATT	188	55.01	1.67	0.00	0.00
SAFE SPCATT	188	55.02	1.67	0.00	0.00
SAFE SPE	188	16.77	0.51	0.25	2.66
SMILES APE	205	27.88	0.88	43.24	60.49
SMILES BPE	205	25.05	0.79	0.00	0.00
SMILES SBPE	205	49.13	1.56	0.00	0.00
SMILES SMIRK	205	67.09	2.12	0.00	0.00
SMILES SMIRK SPCATT	205	63.68	2.02	0.00	0.00
SMILES SPCATT	205	49.16	1.56	0.00	0.00
SMILES SPE	205	16.55	0.52	19.72	58.05

Table 22: SMILES and SAFE tokenizers on the BBBP validation inputs (205 attempted rows).

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
TSDY BRICS	205	11.10	0.35	14.51	93.17
TSDY MMPA	205	23.40	0.74	5.90	80.98
TSID BRICS	205	11.10	0.35	18.77	98.05
TSID MMPA	205	23.40	0.74	9.72	95.61
TSIS BRICS	205	7.32	0.23	27.78	98.05
TSIS MMPA	205	18.00	0.57	12.50	95.61
TSSA BRICS	205	16.54	0.52	8.64	88.78
TSSA MMPA	205	31.79	1.01	2.72	65.37
Enhanced TSDY	205	39.33	1.25	0.79	26.83
Base TSDY	205	28.86	0.91	4.68	77.56
Base TSID	205	28.86	0.91	6.34	92.20
Base TSIS	205	22.73	0.72	8.05	92.20
Base fragSMILES	205	25.60	0.81	10.67	60.98
PS-TSDY	205	17.20	0.54	9.92	87.32
PS-TSID	205	17.20	0.54	16.53	95.61
PS-TSIS	205	12.75	0.40	22.31	95.61
PS-fragSMILES	205	20.79	0.66	12.86	61.95

Table 23: Fragment tokenizers on the BBBP validation inputs (205 attempted rows).

BBBP test

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
SAFE APE	163	24.72	0.86	18.52	100.00
SAFE BPE	163	13.61	0.47	0.00	0.00
SAFE SBPE	163	44.57	1.54	0.00	0.00
SAFE SMIRK	163	60.46	2.10	0.00	0.00
SAFE SMIRK GPE	163	13.61	0.47	0.00	0.00
SAFE SMIRK SGPE	163	44.57	1.54	0.00	0.00
SAFE SMIRK SPCATT	163	44.61	1.55	0.00	0.00
SAFE SPCATT	163	44.61	1.55	0.00	0.00
SAFE SPE	163	13.11	0.45	0.37	4.29
SMILES APE	194	19.45	0.72	37.92	48.97
SMILES BPE	194	17.23	0.64	0.00	0.00
SMILES SBPE	194	36.35	1.35	0.00	0.00
SMILES SMIRK	194	52.72	1.96	0.00	0.00
SMILES SMIRK SPCATT	194	48.41	1.80	0.00	0.00
SMILES SPCATT	194	36.39	1.35	0.00	0.00
SMILES SPE	194	12.08	0.45	15.88	45.88

Table 24: SMILES and SAFE tokenizers on the BBBP test inputs (194 attempted rows).

Tokenizer	Encoded rows	Mean tokens used	Fertility	UNK token rate (%)	Molecules with UNK (%)
TSDY BRICS	194	8.52	0.32	16.58	93.81
TSDY MMPA	194	19.62	0.73	5.78	73.71
TSID BRICS	194	8.52	0.32	19.18	98.45
TSID MMPA	194	19.62	0.73	9.32	94.33
TSIS BRICS	194	5.01	0.19	31.28	98.45
TSIS MMPA	194	14.97	0.56	12.09	94.33
TSSA BRICS	194	12.23	0.46	9.40	86.08
TSSA MMPA	194	26.36	0.98	2.17	45.36
Enhanced TSDY	194	28.30	1.05	0.71	19.07
Base TSDY	194	23.44	0.87	4.62	67.53
Base TSID	194	23.44	0.87	6.55	89.18
Base TSIS	194	18.31	0.68	8.39	89.18
Base fragSMILES	194	19.21	0.71	8.77	47.42
PS-TSDY	194	13.32	0.50	10.57	86.08
PS-TSID	194	13.32	0.50	15.02	93.81
PS-TSIS	194	9.47	0.35	21.11	93.81
PS-fragSMILES	194	14.04	0.52	11.57	50.00

Table 25: Fragment tokenizers on the BBBP test inputs (194 attempted rows).